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**DEPARTMENT OF INFORMATION
TECHNOLOGY**



**Secure Patient Data & Insurance Management
System**

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3.6 Conceptual Models -Event Table, Object Diagram:

3.6.1 Event Table:

Introduction:

An Event Table is a software engineering artifact that identifies the significant events occurring during the execution of the system. It describes the interactions between different users and the application, the actions that trigger these interactions, and the corresponding system responses. The Event Table provides a structured view of the dynamic behavior of the system and helps in understanding how business processes are executed.

In the Secure Patient Data & Insurance Management System, multiple users such as Administrator, Hospital Department, Medical Department, Insurance Company, and Patients interact with the system based on their roles and permissions. These interactions include patient registration, medical record management, bill generation, insurance claim verification, appointment management, and report generation.

Each event represents a specific business activity that is initiated by a user and processed by the system according to predefined business rules. The Event Table ensures that every user action produces an expected system response, making the application easier to analyze, develop, test, and maintain.

The following table summarizes the major events supported by the system.

Event ID	Event	Trigger	Actor	System Response
E01	User Login	User enters valid username and password	Admin / Hospital Staff / Insurance Company	System authenticates the user and redirects to the appropriate dashboard.
E02	Patient Registration	Registration form is submitted	Hospital Department	System validates the information, generates a unique Patient ID, and stores patient details in the database.
E03	Update Patient Information	Patient details are modified	Hospital Department	System validates and updates the patient's information successfully.
E04	Add Medical Record	Doctor submits diagnosis and treatment details	Medical Department	Medical record is stored and linked with the corresponding Patient ID.
E05	Upload Medical Bill	Hospital uploads	Medical Department	Bill is saved in the database and

		treatment bill		becomes available for insurance verification.
E06	View Patient History	Patient ID is searched	Hospital Department	System retrieves complete medical history, reports, and bills.
E07	Upload Medical Report	Lab report is uploaded	Medical Department	Medical report is securely stored and linked with the patient's profile.
E08	Search Patient	Patient ID or Name is entered	Hospital Department / Insurance Company	System displays the patient's information and associated medical records.
E09	Submit Insurance Claim	Insurance claim request is submitted	Insurance Company	System retrieves all medical bills and reports for claim verification.
E10	Verify Insurance Claim	Retrieved documents are verified	Insurance Company	System validates the uploaded

				records and updates the claim status.
E11	Generate Reports	User clicks Generate Report	Administrator	System generates patient or insurance reports in PDF format.
E12	User Logout	User clicks Logout	All Users	System ends the current session and redirects the user to the login page.

3.6.2 Object Diagram:

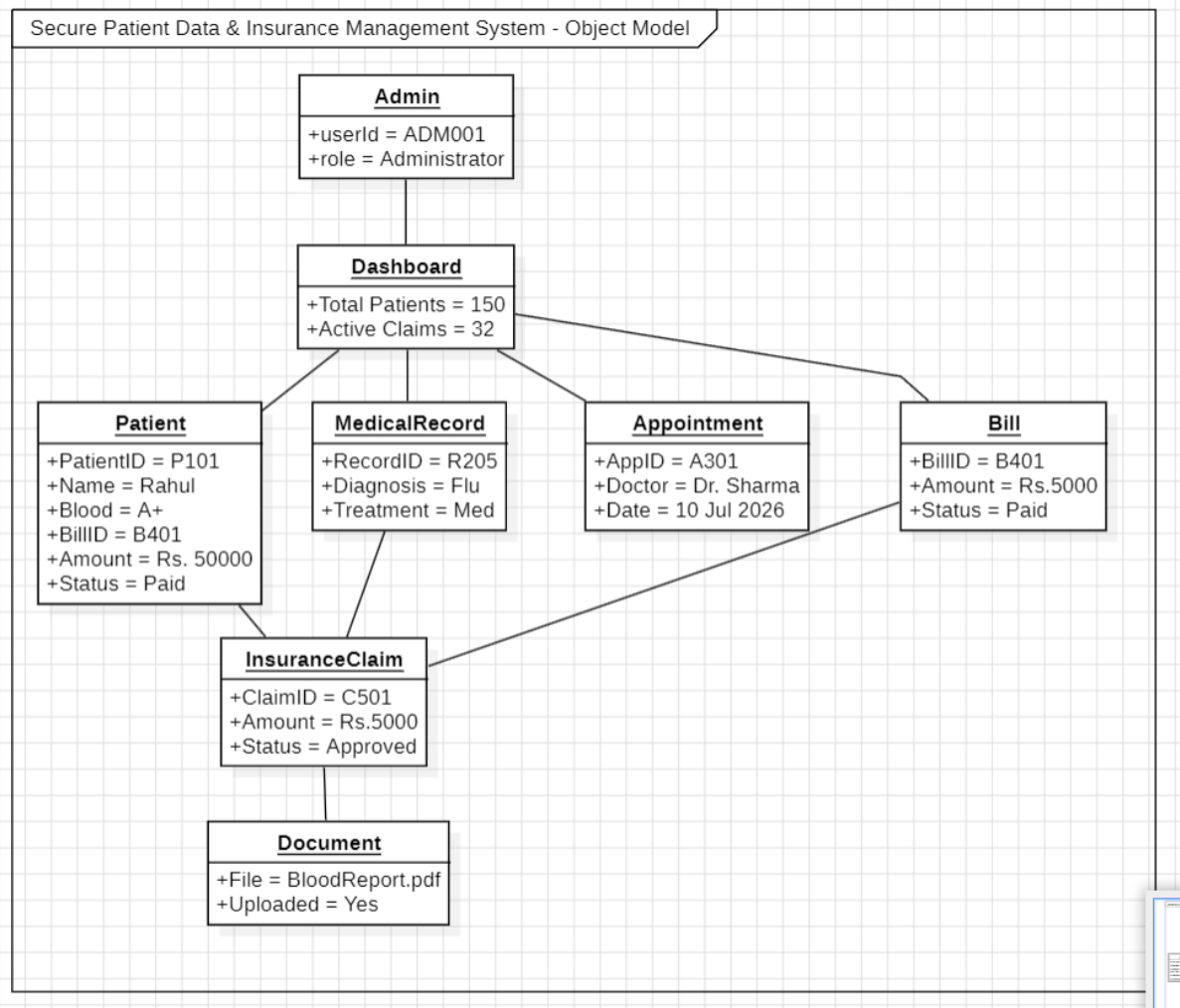
Introduction:

An Object Diagram is a UML (Unified Modeling Language) structural diagram that represents the instances (objects) of classes at a specific point in time. Unlike a Class Diagram, which describes the structure of the system, an Object Diagram shows the actual objects that exist during the execution of the application along with their current attribute values and relationships.

The Object Diagram provides a snapshot of the system by illustrating how different objects interact with each other in a real-world scenario. It helps developers, testers, and stakeholders understand the runtime behavior of the system and verify whether the designed class relationships work correctly during execution.

In the Secure Patient Data & Insurance Management System, the Object Diagram represents real instances of entities such as Administrator, Patient, Medical Record, Medical Bill, Insurance Claim, and Medical Report. It demonstrates how a patient is registered, how medical records and bills are linked with the patient, and how the insurance company accesses these records for claim verification.

The Object Diagram is useful for validating object relationships, understanding the system's execution flow, and ensuring that the implemented application behaves according to the project requirements. It also serves as a valuable reference during testing and debugging by showing actual data stored in different objects.



Relationships:

- Admin monitors the Dashboard.
- Dashboard displays Patient, Medical Record, Appointment, and Bill information.
- The Patient is linked to a Medical Record, Appointment, and Bill.
- The Bill is used to create an Insurance Claim.
- The Insurance Claim is supported by uploaded Documents.